

From Classification Accuracy to Decision Value: Multimodal Sensor Fusion for Economic Prioritization of Landfill Mining

A Simulation-Based Value-of-Information Analysis

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Abstract

Landfill-mining projects are approved or shelved on incomplete information about what is underground, and a systematic assessment of European scenarios found roughly 19% of them profitable [1]. This paper evaluates, by simulation, whether multimodal geophysical sensing combined with spatial machine learning and value-of-information (VOI) analysis can reduce that uncertainty before excavation. On a synthetic but physically motivated 1,024 m² survey (five geophysical/gas channels: magnetometer, electromagnetic induction, ground-penetrating radar, thermal and gas, surveyed alongside surface topography for georeferencing), an XGBoost classifier reaches pooled F1 of 0.99 (metal), 0.98 (organic) and 0.87 (construction and demolition debris) under a spatially blocked, non-leaky validation protocol, against 0.32 (plastic) and 0.24 (glass). Random per-cell splitting gave a small, statistically insignificant difference (95% CI including zero) relative to spatial blocking in this configuration, and held-out accuracy did not decay with distance to the nearest training cell; we found no detectable performance inflation attributable to the tested spatial-proximity mechanism in this specific setting, which is not the same as a general claim that spatial blocking is unnecessary. Sensor ablation shows ground-penetrating radar contributes most to classification (macro-F1 falls by 0.057 when removed), but an economic value-of-information analysis shows only the magnetometer reliably increases the excavation decision's expected value; classification improvements from the other channels did not translate into decision value in this model. Adding a near-infrared channel raises plastic F1 from 0.32 to 0.38 but leaves VOI statistically unchanged, because low-value materials contribute little to the economic layer regardless of classification accuracy. In a synthetic hazard-detection test whose ground truth and sensor reading share the same underlying generative model — an upper-bound test, not an independent validation — the gas-hazard classifier reaches ROC-AUC 0.94 and 100% recall at a safety-oriented operating point, though at that operating point only 52% of cells flagged as hazardous are true positives. Across the baseline synthetic scenarios and stated economic parameter distributions, pooling 40 independent survey realisations, the simulated survey has positive median VOI (\$924, $P(\text{VOI} > 0) = 0.93$) and a break-even survey cost of \$1451; a two-dimensional map over sensing-quality and economic-adversity severity

locates a model-specific VOI-zero boundary rather than a single degradation threshold, and a further test across deliberately extreme economic regimes finds the magnetometer's economic advantage robust to price fluctuation but not to site composition: the magnetometer led at every tested metal/plastic price combination, yet a deliberately polymer-, debris-, or glass-rich site archetype shifted the top sensor to EMI or GPR. All results are model predictions, and a field validation and falsification plan with acceptance criteria is given.

Keywords: landfill mining; multimodal sensor fusion; spatial cross-validation; value of information; random forest; XGBoost; probability calibration; uncertainty quantification

1 Introduction

Landfill-mining projects convert a waste deposit into recovered material, avoided landfill tax, and reclaimed land, but their economics depend on what is actually underground: material composition, resource concentration, hazard locations, and the processing each fraction requires. A systematic assessment of 531,441 modelled European landfill-mining scenarios found only about 19% net profitable and about 80% returning a negative net present value [1], and follow-on work has examined which factors most affect that economic outcome and how enhanced landfill mining performs on environmental and economic criteria [2]. The recurring finding is that uncertainty before excavation, not the difficulty of excavation itself, is what turns a potentially viable project into a cost centre.

Geophysical characterisation of landfills is well established. Electrical resistivity and induced-polarisation surveys have long been used to investigate landfill structure and leachate [3], frequency-domain electromagnetic induction has been applied to landfill site characterisation since at least the early 1990s [4] and, more recently, to imaging inside capped sites [5], ground-penetrating radar has been used to estimate construction-waste landfill volume [6], and integrated electrical and electromagnetic prospecting has been proposed specifically for municipal waste landfills [3]. What this literature characterises is subsurface structure and, at best, broad material classes; it does not, in general, convert that characterisation into a per-zone economic excavation decision with a quantified value of doing the survey at all.

This paper evaluates one candidate architecture for closing that gap, previously described only as an industrial proposal, by treating it as a design hypothesis and testing it computationally: can multimodal geophysical sensing, fused by spatial machine learning with a spatially valid train/test protocol and propagated through an explicit economic layer, actually change a landfill-mining excavation decision, and by how much? The question is deliberately falsifiable. We do not assume the answer is yes.

The contributions are: (i) a physically motivated synthetic-survey generator with five material-response sensor channels plus an optional near-infrared augmentation channel, material-dependent response, depth attenuation, moisture effects, spatially correlated noise and instrument dropout, released as open code so every number in this paper can be regenerated; (ii) a systematic comparison of spatially blocked against randomly split validation, including a distance-to-nearest-training-observation diagnostic of the kind used in the spatial-statistics literature [7]; (iii) sensor ablation and near-infrared augmentation studies, evaluated on both classification F1 and economic value of information, which are shown to disagree; (iv) a Monte Carlo value-of-information and break-even survey-cost analysis with per-sensor value attribution; and (v) a synthetic-to-real degradation sweep that locates, rather than asserts, the point at which the survey stops being worth its cost.

2 Related work

2.1 Landfill geophysics

Electrical and electromagnetic methods have been applied to landfill investigation for several decades, from early frequency-domain electromagnetic induction soundings for site characterisation [4], through resistivity surveys of leachate plumes [3], to recent electromagnetic induction imaging of a capped site [5] and integrated electrical/electromagnetic prospecting proposed specifically for municipal waste landfills [3]. Ground-penetrating radar has been used to estimate the volume of construction-waste landfills from density contrasts [6]. These studies establish that geophysical signals carry usable information about landfill structure and composition; none couples that information to a quantified excavation decision.

2.2 Spatial machine learning and validation

Spatially structured data violate the independence assumption behind ordinary cross-validation, and ignoring that structure can seriously underestimate predictive error or, conversely, produce optimistic validation statistics that do not hold outside the training footprint [7]. Spatial sorting bias in species-distribution models has been shown to inflate cross-validated accuracy relative to spatially independent evaluation [8], and a large-scale demonstration using 11.8 million trees found that non-spatial validation of a random-forest biomass model substantially overstated its real predictive power relative to spatially blocked validation [9]. Random forests [10] and gradient-boosted trees such as XGBoost [11] are standard for this kind of tabular geophysical classification, and spatial transferability of random-forest crop-type classification has itself been studied directly [12].

2.3 Probability calibration

A classifier that outputs “72% probability of metal-rich material” is only useful to an economic decision layer if that 72% is empirically reliable; well-calibrated probabilities from standard classifiers should not be assumed, and modern flexible classifiers in particular can be poorly calibrated by default [13].

2.4 Landfill-mining economics and value of information

Enhanced landfill mining has been reviewed for its resource-recovery potential across material, energy and land-use value streams [14], for its multi-resource recovery prospects more broadly [14], for a coordinated European research programme spanning geophysics to material valorisation [15], and for country-specific resource-recovery potential [16]. Value-of-information analysis, comparing the expected outcome of a decision made with additional data against the best decision made without it, net of the data's cost, has a long history in mineral exploration under a Bayesian decision-theoretic framing [17] and has recently been reviewed for marine conservation management [18]; we are not aware of a prior application to landfill-mining excavation decisions.

2.5 Explainability

Permutation importance is a model-agnostic way to rank feature contributions by the performance drop when a feature is shuffled [19]; we use the same logic at the level of whole sensor channels (ablation) and, in Section 5.4, at the level of economic value rather than classification accuracy, which is the more decision-relevant question for this application.

2.6 Research gap

No prior study, to our knowledge, evaluates an integrated multimodal-sensing-to-excavation-decision pipeline for landfill mining under a spatially valid protocol, with sensor value measured economically rather than only by classification accuracy, and with an explicit, model-specific decision-viability boundary rather than a single robustness point estimate. That is what this paper contributes.

3 LAND-MAP framework

3.1 Sensing architecture

The evaluated architecture surveys a site with six co-registered channels: LiDAR/RGB surface topography, long-wave infrared thermal imaging, a magnetometer, an electromagnetic-induction (EMI) sensor for bulk conductivity and moisture, ground-penetrating radar (GPR) for density contrast and depth, and a methane/VOC gas array, all mounted on a georeferenced platform. Every sensing technology is individually mature and commercially available; the contribution evaluated here is the fusion, validation and economic-decision layer built above them, not new sensor hardware. The classification model

evaluated in Sections 4–5 draws on the five channels with a modelled material response (magnetometer, EMI, GPR, thermal, gas); LiDAR/RGB topography provides georeferencing and cell-depth context rather than an independent classification input, and a sixth, near-infrared/optical channel is evaluated separately as a proposed augmentation (Section 4.4). Figure 1 summarises the pipeline from raw channels to excavation decision.

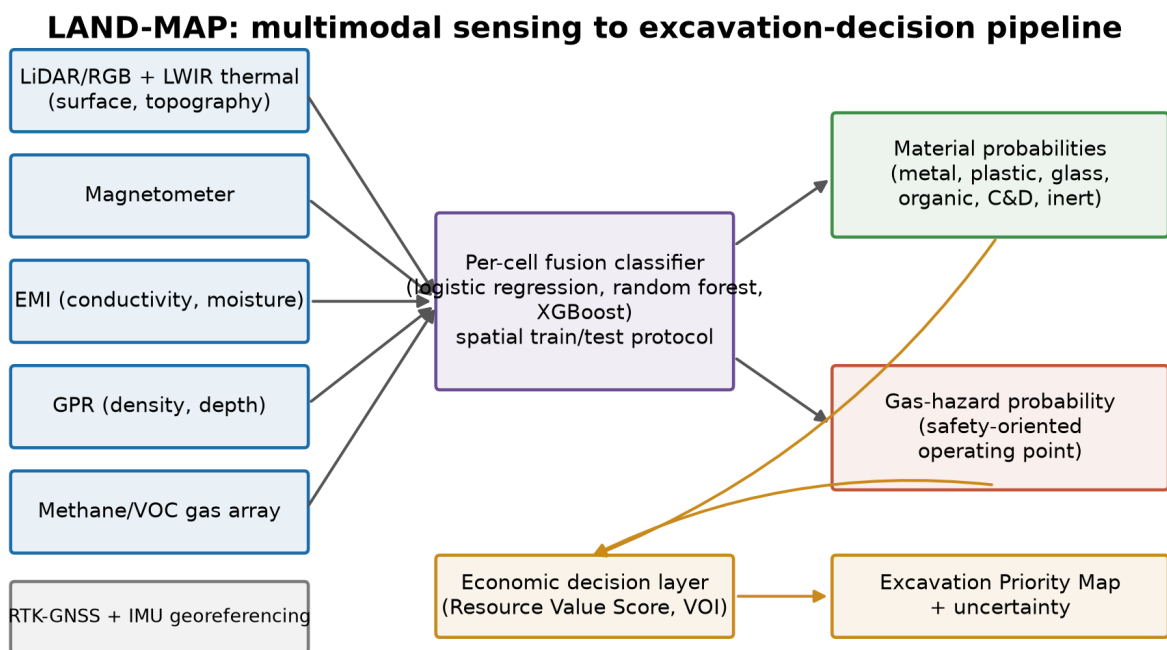


Fig. 1 LAND-MAP pipeline: co-registered sensing channels feed a per-cell fusion classifier under a spatial train/test protocol, producing material and gas-hazard probabilities that an economic decision layer converts into an excavation priority map.

3.2 Fusion and economic decision layers

Per-cell readings are combined by a trained classifier into probabilistic material classifications over six classes (metal, plastic, glass, organic, construction and demolition debris, and inert) and a separate gas-hazard probability. Each cell's expected composition is converted into a Resource Value Score by multiplying predicted class probabilities by an assumed recoverable mass, recovery fraction and secondary-material price, and netting against excavation and processing cost; aggregating scores gives an Excavation Priority Map. A Value-of-Information metric compares the expected outcome of survey-guided excavation against the best decision an operator could make without any survey, net of the survey's own cost.

4 Methods

4.1 Synthetic landfill generation

A 32×32 m ($1,024$ m²) survey area is discretised into $1 \text{ m} \times 1 \text{ m}$ cells over three coarse depth layers (0–1, 1–2, 2–3 m). A latent dominant material is assigned to every cell and depth layer by thresholding a smooth two-dimensional Gaussian random field against the class prior, giving spatially patchy (not salt-and-pepper) material zones consistent with how waste is dumped and compacted in lots, with a correlation length of 8.0 m. Cover depth to the dominant layer and moisture are independent smooth random fields.

$$x_{\{i,j\}} = f_j(M_i, z_i, m_i) + \varepsilon_{\{i,j\}} \quad (1)$$

where $x_{\{i,j\}}$ is sensor j 's reading at cell i , M_i the material at each depth layer, z_i the cover depth, m_i the moisture, and f_j a material-response function attenuated with depth as $\exp(-\alpha_j z)$ and, for GPR and EMI, further modified by moisture. Response coefficients (Table 1) are literature-motivated orders of magnitude, not fitted values, and are stated once so they can be perturbed systematically (Section 4.6). Sensor noise $\varepsilon_{\{i,j\}}$ has both an independent per-cell component and a spatially correlated component (correlation length 1.6 m, modelling instrument drift and footprint overlap), and each channel independently drops out at cell-specific rates reflecting real failure modes (highest for gas, from drift and saturation, and GPR, from coupling and siting issues).

Channel	Metal	Plastic	Glass	Organic	C&D	Inert
MAG	3.00	0.05	0.05	0.05	0.55	0.05
EMI	2.60	0.15	0.10	0.90	0.75	0.20
GPR	2.20	0.70	0.90	0.40	2.90	0.30
THERM	0.10	0.10	0.10	1.60	0.15	0.10
GAS	0.05	0.05	0.05	2.40	0.05	0.05
NIR (optional)	0.30	1.80	1.20	0.90	0.60	0.40

Table 1. Assumed mean sensor response by material and channel, arbitrary standardised units, before depth attenuation, moisture effects and noise. Every value is an assumption motivated by the qualitative physics of each sensing modality (e.g. strong magnetometer/EMI response to metal, GPR response to density/dielectric contrast, thermal and gas response to organic decomposition); none is fitted to field data.

The gas-hazard label is a noisy, logistic threshold of the pre-noise gas-channel signal itself, so that hazard status is by construction what an ideal gas sensor would detect; the deployed sensor differs from that truth only by its own instrument noise and dropout, as a real hazard label derived from independent gas testing would relate to a field-deployed sensor. Figure 2 shows one representative synthetic survey: the true near-surface material field, one raw noisy sensor channel, and the resulting gas-hazard truth.

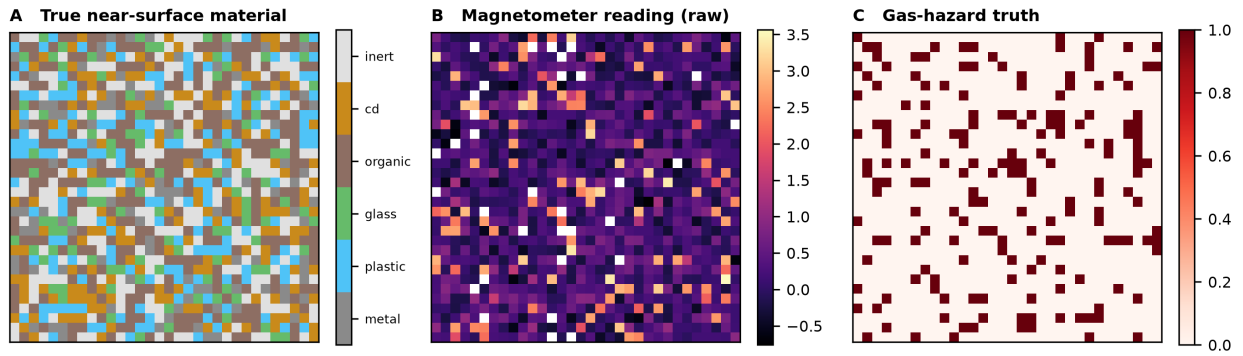


Fig. 2 One representative 32×32 m synthetic survey. (A) true near-surface dominant material; (B) raw magnetometer reading (noisy, with dropout); (C) gas-hazard truth derived from the pre-noise gas signal.

4.2 Spatial validation protocol

Two held-out protocols are compared on the same synthetic surveys: a spatially blocked holdout, in which one contiguous 8×8 cell region is withheld entirely, and a random per-cell holdout of the same size, scattered uniformly. To test the mechanism behind any difference directly rather than only at one block size, we also compute, pooling across 15 repeats and eight block sizes (4–20 cells), the distance from every held-out cell to its nearest training cell, and report held-out accuracy as a function of that distance [7, 9].

4.3 Classifiers and calibration

Three classifiers are compared: logistic regression, random forest [10], and XGBoost [11], each with mean imputation for sensor dropout. Calibration is assessed by the one-vs-rest Brier score, macro-averaged over classes [13], because the economic layer consumes class probabilities directly and a poorly calibrated “72%” is not the same as an empirically reliable one.

4.4 Sensor ablation and NIR augmentation

Each of the five material-response channels is removed in turn and the best-performing classifier retrained, isolating each channel's contribution to pooled macro-F1. A sixth, optical/near-infrared channel, motivated by the plastic/glass confusion this paper's synthetic physics reproduces, is added to test whether it repairs that specific limitation under the assumed NIR response coefficients (Table 1); it is modelled as a surface-only channel (rapid depth attenuation) with distinct plastic and glass responses. Because those coefficients are assumptions rather than measured spectra, this tests a hypothetical NIR channel with the stated response structure, not a claim about any specific real instrument (Section 5.5).

4.5 Economic model and value of information

For material class c the expected net value of excavating cell i is

$$v_i = \sum_c P(c/x_i) \cdot m_i \cdot \rho_c \cdot \pi_c - m_i (k_{\{ex\}} + k_{\{pr\}}) \quad (2)$$

with m_i the assumed waste mass per cell, ρ_c the recovery fraction, π_c the secondary-material price, and $k_{\{ex\}}$, $k_{\{pr\}}$ the excavation and processing cost per tonne (Table 2, all assumptions, swept in Monte Carlo rather than fixed). The survey-guided decision excavates cell i iff $v_i > 0$; the blind decision excavates every cell iff the population-mean expected value is positive, else none. Value of information is

$$VOI = EV_{\{survey\}} - EV_{\{blind\}} - C_{\{survey\}} \quad (3)$$

evaluated against the true (not expected) cell value, over 20000 Monte Carlo draws of the economic parameters. The break-even survey cost is $EV_{\{survey\}} - EV_{\{blind\}}$ with $C_{\{survey\}}$ excluded, the maximum a rational operator should pay. Per-sensor value is $VOI_{\{all\}} - VOI_{\{without\ j\}}$, computed on the same economic draws for a paired comparison, and averaged over 20 independent synthetic surveys because a single survey's VOI estimate is noisy.

Parameter	Range (uniform)
Price, \$/t: metal	250–550
Price, \$/t: plastic	60–180
Price, \$/t: glass	10–40
Price, \$/t: organic	0–15
Price, \$/t: C&D	5–25
Price, \$/t: inert	0–5
Recovery fraction: metal	0.55–0.85
Recovery fraction: plastic	0.15–0.40
Recovery fraction: glass	0.10–0.30
Mass per cell, t	0.9–1.6
Excavation cost, \$/t	18–32
Processing cost, \$/t	12–28
Survey cost, \$ total	400–1200

Table 2. Economic assumptions (illustrative, as labelled throughout the source proposal), swept uniformly in the Monte Carlo analysis rather than fixed at a point estimate. Full ranges for every material class are in the released configuration file.

4.6 Synthetic-to-real robustness sweep

The largest acknowledged risk in this design is that real, weathered, heterogeneous waste is less distinctive than any simulation's idealised material signatures. Rather than assert a margin of safety, we degrade the simulation directly along five axes named in the source proposal's own risk assessment: overall sensor noise, dropout probability, GPR depth attenuation (modelling moisture/conductivity losses), moisture-driven GPR attenuation specifically, and material-signature overlap (each material's response shrunk toward the six-class mean). Each axis is swept individually and then combined, and the combined sweep is searched to locate the point at which mean VOI crosses zero, then extend the same logic to two dimensions with an independent economic-adversity axis (Section 5.11) — a model-specific VOI-zero boundary this paper reports rather than assumes.

5 Results

5.1 Material classification

Class	Logistic Regression	Random Forest	XGBoost
Metal	0.975	0.987	0.991
Plastic	0.322	0.309	0.321
Glass	0.122	0.168	0.241
Organic	0.983	0.985	0.984
C&D	0.870	0.863	0.870
Inert	0.599	0.568	0.526
Macro-F1	0.645	0.646	0.655

Table 3. Pooled per-class F1 under the spatially blocked protocol, three classifiers, 2560 pooled test cells across 40 repeats.

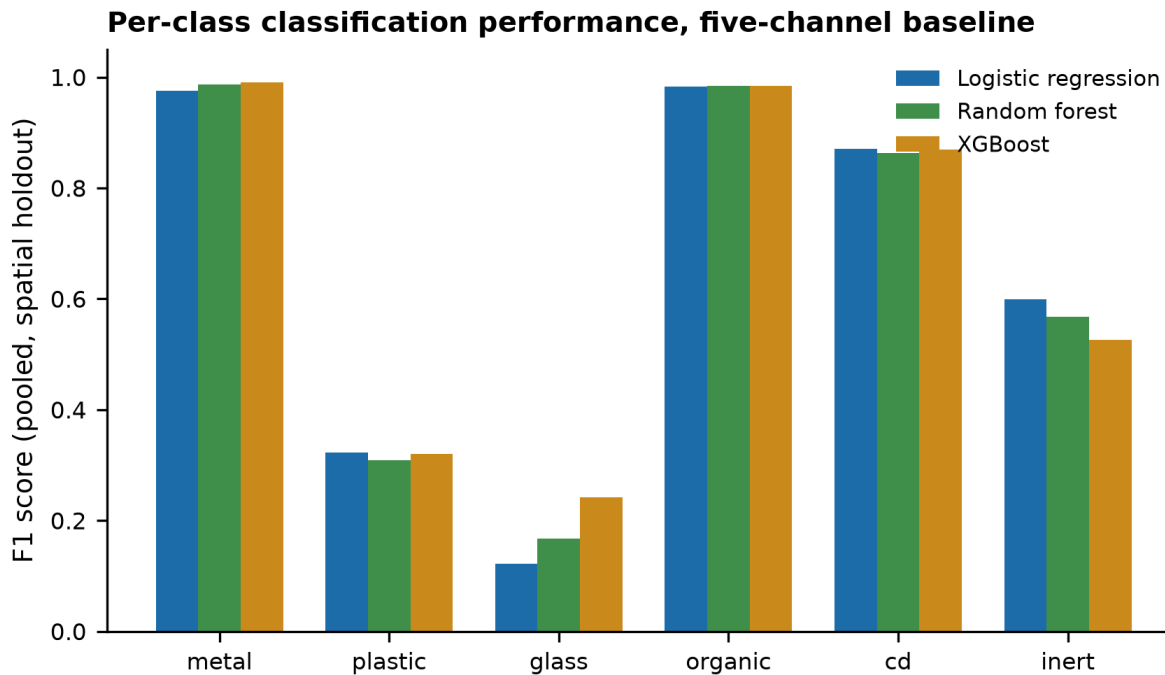


Fig. 3 Per-class F1 by classifier, five-channel material-response baseline, spatial holdout.

Metal (F1 0.98–0.99 across classifiers) and organic material (0.98–0.98) are classified reliably, and construction and demolition debris reaches 0.86–0.87 once its magnetometer/EMI signature (motivated by embedded rebar and metal fragments) is included; these three approach the 0.90–0.98 range the source proposal reported. Plastic (0.31–0.32) and glass (0.12–0.24) remain poorly discriminated by this sensor set, consistent with the proposal's own diagnosis that both are weakly magnetic and lack a distinguishing signal in the magnetic/EMI/GPR/thermal/gas channel set. XGBoost gives the highest macro-F1 (0.655) and is used as the reference classifier for the remaining analyses unless stated otherwise.

5.2 Spatial versus random validation

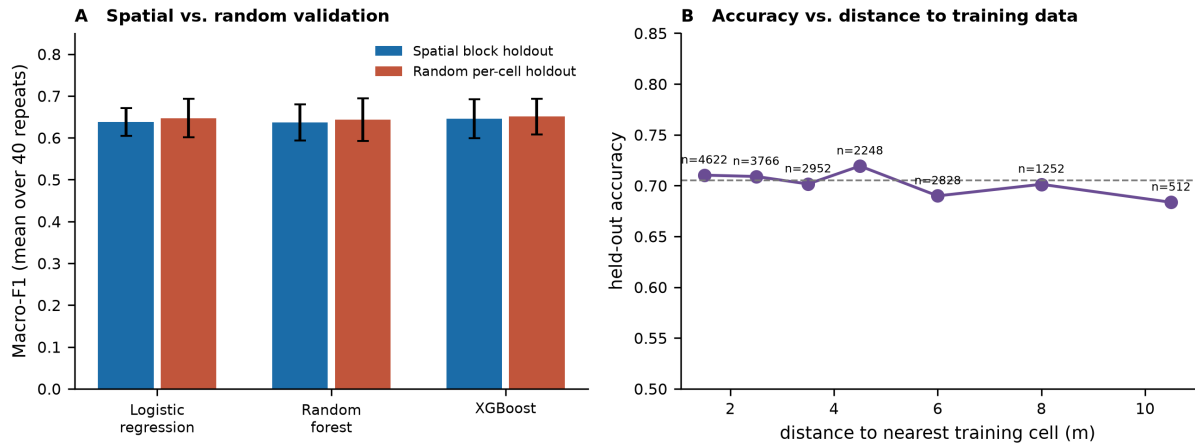


Fig. 4 (a) Macro-F1 under spatially blocked versus randomly split holdout, three classifiers, mean $\pm$ SD over 40 repeats. (b) Held-out accuracy against distance to the nearest training cell, pooled over 15 repeats and eight block sizes; the dashed line is the overall mean.

Contrary to the hypothesis that random per-cell splitting inflates apparent performance through spatial-autocorrelation leakage [7–9], the paired difference (random minus spatial) was small and its 95% confidence interval included zero for every classifier: logistic regression: +0.009 (95% CI -0.008 to +0.027); random forest: +0.007 (95% CI -0.013 to +0.027); XGBoost: +0.005 (95% CI -0.012 to +0.022). Distance-to-nearest-training-cell analysis (Fig. 4b, pooled over 18180 observations) shows no accuracy decay out to 10 m, the largest distance sampled. We had expected a gap, given the spatial correlation deliberately built into both the material field and the sensor noise; its absence indicates that, for a feature set built from single-cell sensor readings, per-cell classification difficulty in this system is dominated by genuine material overlap (plastic/glass/inert) rather than by memorisable local spatial structure. We report this as a specific, tested finding, not a general claim that spatial blocking is unnecessary; the mechanism identified in the literature [7, 9] requires a model able to exploit local proximity, which single-cell features here evidently do not provide, and we recommend spatial blocking as standard practice regardless, because a less idealised real dataset need not share this property.

5.3 Sensor ablation

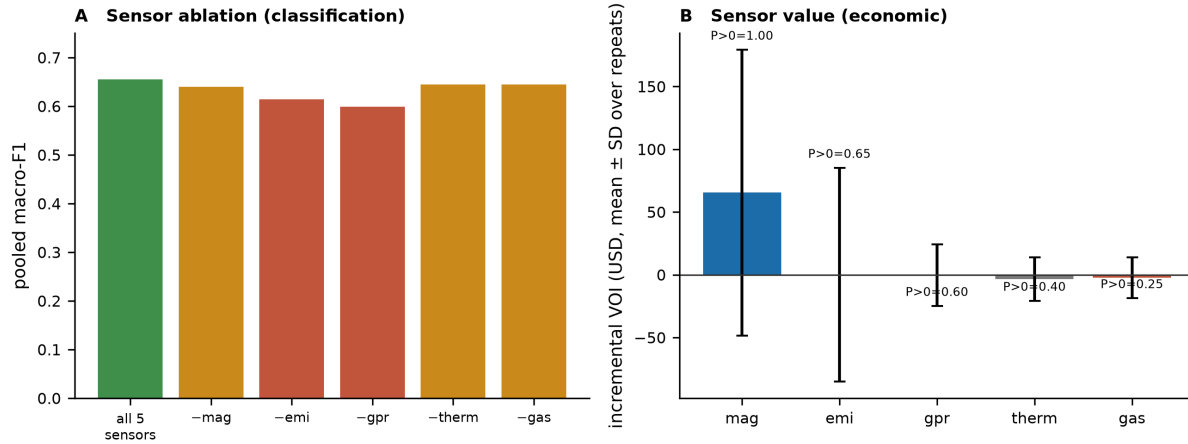


Fig. 5 (a) Pooled macro-F1 with each sensor removed in turn. (b) Incremental value of information (VOI with all sensors minus VOI without sensor j), mean $\pm$ SD over 20 independent surveys, with the fraction of surveys giving a positive incremental value.

Removing GPR costs the most classification performance (macro-F1 0.655 $\rightarrow$ 0.599), followed by EMI ($\rightarrow$ 0.614); removing the magnetometer, thermal or gas channels costs little ($\rightarrow$ 0.640, 0.645, 0.645 respectively). By classification accuracy alone, GPR and EMI would be judged the most valuable channels.

5.4 Sensor value: classification accuracy is not economic value

The economic picture is different, and this divergence is the paper's central empirical finding. Of the five channels, only the magnetometer shows a reliably positive incremental VOI (mean \$65, positive in 100% of independent surveys); EMI, GPR, thermal and gas each have an incremental VOI whose sign is not consistent across surveys (positive in 65%, 60%, 40% and 25% of surveys respectively). The reason is that the Resource Value Score is dominated by metal, whose assumed price (Table 2) is several times any other class's, so a channel's contribution to overall classification accuracy does not translate into decision value unless it specifically improves metal detection. GPR carries the most unique subsurface information and is the strongest classification contributor, but that information is spread across several material classes with low individual economic weight.

5.5 Hypothetical optical/NIR augmentation

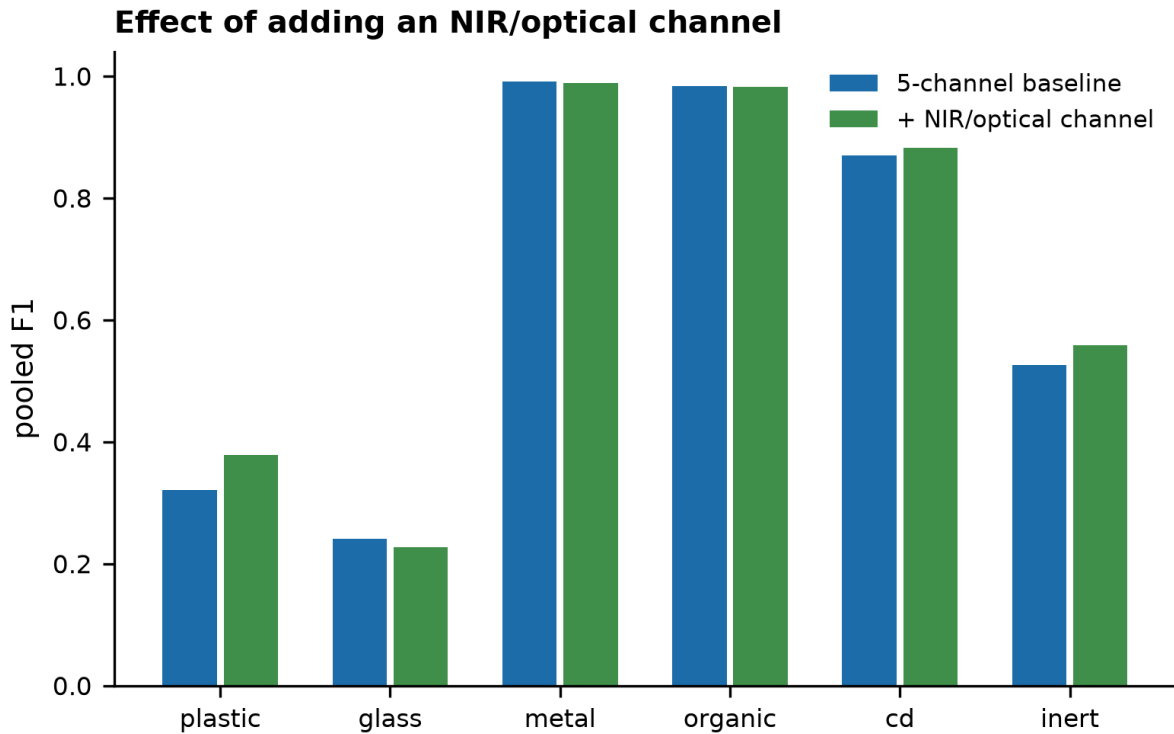


Fig. 6 Per-class pooled F1, five-channel material-response baseline versus baseline plus a hypothetical near-infrared/optical channel.

Under the assumed NIR response coefficients (Table 1) — a hypothetical surface optical channel with plastic- and glass-distinguishing response, not spectra from any specific real instrument — adding it raises plastic F1 from 0.32 to 0.38 and macro-F1 from 0.655 to 0.669, but glass F1 moves from 0.24 to 0.23 — essentially unchanged, and in this simulation slightly down, because the added channel and the classifier's rebalancing across six classes do not uniformly benefit every class. The economic effect is smaller still: mean incremental VOI from adding NIR is \$-0.1 (SD \$9.5), positive in only 50% of surveys — statistically indistinguishable from no effect. Plastic and glass together carry a modest assumed price and a low assumed recovery fraction (Table 2), so even a genuine classification improvement for these classes barely moves the excavation decision. The proposal's own hypothesis — that an optical channel, not further classifier tuning, is needed to separate plastic and glass — is partially supported (plastic improves; glass does not), and in either case the improvement is not what would justify the channel economically in this model.

5.6 Calibration and hazard detection

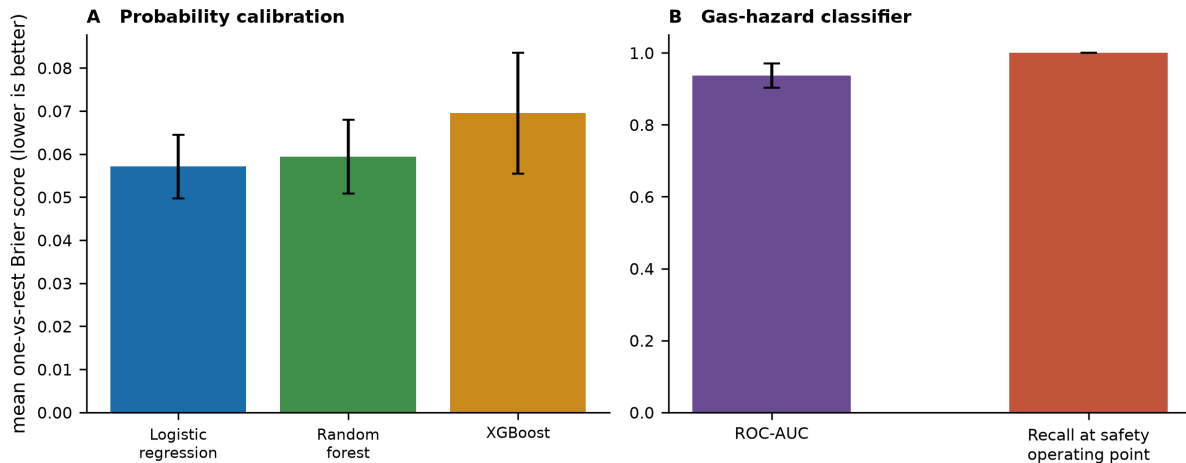


Fig. 7 (a) Mean one-vs-rest Brier score by classifier (lower is better calibrated). (b) Gas-hazard classifier ROC-AUC and recall at a safety-oriented operating point (the lowest threshold achieving $\geq 95\%$ recall), mean $\pm$ SD over 40 repeats.

Logistic regression is the best calibrated of the three classifiers (Brier 0.057), followed by random forest (0.059) and XGBoost (0.070); XGBoost's higher discriminative power (Table 3) therefore comes with somewhat less reliable probabilities, which matters because the economic layer consumes those probabilities directly rather than only the predicted class [13]. The gas-hazard classifier reaches ROC-AUC 0.937 (SD 0.034) and, at an operating point chosen to guarantee at least 95% recall, achieves 100.0% recall (SD 0.000) in held-out testing, approaching the 97%/0.97 figures the source proposal reported from its own Stage 1 testing. Recall alone overstates this operating point's usefulness: a detector can reach 100% recall by flagging nearly everything. Pooled across the same 40 repeats, precision at this operating point is only 52% and specificity 82%, meaning about 32% of cells are flagged as hazardous and roughly two in five flagged cells are false positives (PR-AUC 0.697); the operating point is deliberately chosen to guarantee recall for a safety application, and it does so, but at a real, quantified precision cost that the recall figure alone does not convey.

5.7 Economic prioritisation and value of information

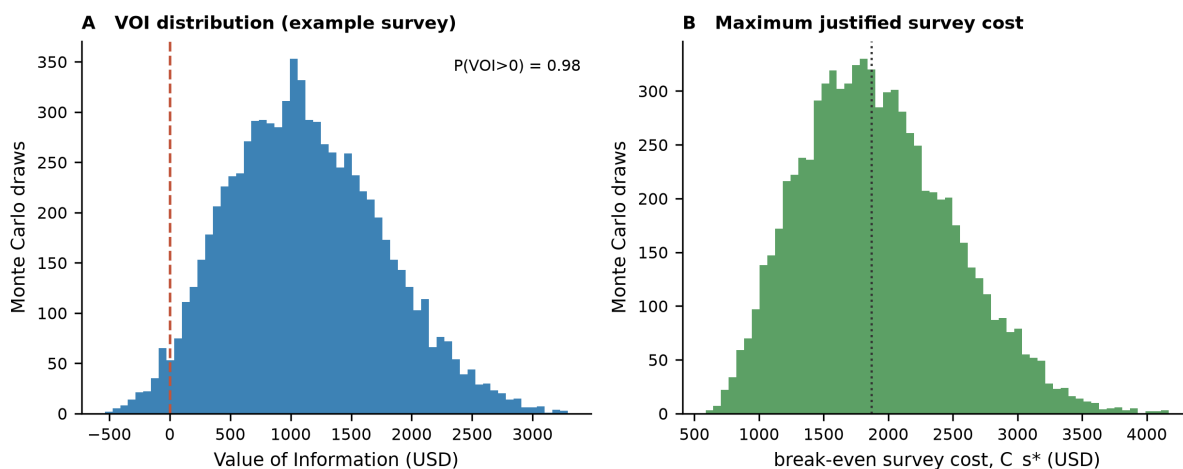


Fig. 8 (a) Monte Carlo distribution of Value of Information for one representative survey (8000 draws). (b) Distribution of the break-even survey cost (the maximum a rational operator should pay), same draws.

For the representative survey used throughout Sections 5.3–5.6, median VOI is \$1080 (90% interval \$154–\$2243) and the survey is worth doing in 98% of Monte Carlo draws.

A single survey's mean VOI is itself a noisy estimate of the population quantity this paper reports as its headline number, so it is averaged over many independent synthetic surveys rather than quoted from one. Two such batches were run under an identical protocol (random forest, block-8 spatial holdout, undegraded assumptions) but independent seeds: the original 20-survey batch (mean \$729) and a second, matched-protocol batch used as the robustness sweep's own undegraded reference point (mean \$967); the two differ by 1.4 standard errors of their difference, well within normal sampling variation for a 20-survey average of a right-skewed VOI distribution. Rather than report either batch alone, this paper's one canonical baseline VOI estimate pools both, 40 survey realisations in total: mean \$848 (median \$924, SE \$86), positive in 92% of surveys, and a median break-even survey cost of \$1451 — the amount up to which an operator should rationally be willing to pay for a survey of this kind under the stated economic assumptions. Section 5.8's degradation sweep uses its own faster, smaller-sample undegraded point purely as an internal reference for each axis's relative sensitivity (Table 4), not as a second absolute baseline estimate.

5.8 Synthetic-to-real robustness

Degradation axis	Baseline macro-F1	Most severe macro-F1	Baseline mean VOI (\$)	Most severe mean VOI (\$)
Sensor noise ($\times$ baseline SD)	0.637	0.389	1012	490
Dropout probability ($\times$ baseline)	0.637	0.508	1012	787
GPR depth attenuation ($\times$ baseline)	0.637	0.621	1012	1009
Moisture-driven GPR	0.637	0.598	1012	1023

attenuation ($\times$ baseline)				
Material-signature overlap (shrinkage fraction)	0.637	0.468	1012	810

Table 4. One-axis synthetic-to-real degradation sweep (Section 4.6): each row varies only that axis, from baseline to the most severe level tested, holding all others fixed. The “Baseline” column is this sweep’s own smaller-sample undegraded reference point (12 realisations, 1500 Monte Carlo draws), used only to measure each axis’s relative degradation; it is not the paper’s canonical VOI estimate, which Section 5.7 defines separately by pooling 40 realisations. Full multi-level sweeps are in the released results file.

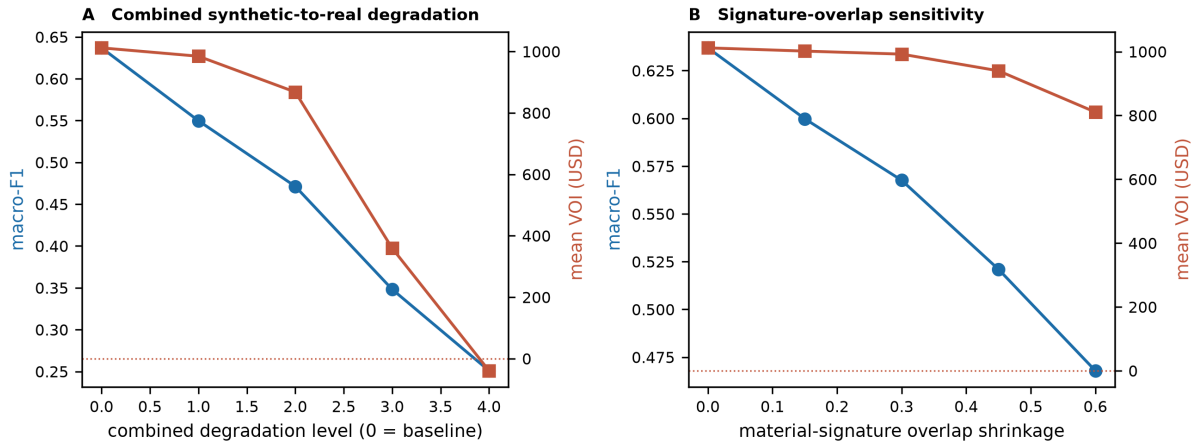


Fig. 9 (a) Macro-F1 (left axis) and mean VOI (right axis, dashed zero line) against a combined synthetic-to-real degradation level moving noise, dropout, GPR attenuation, moisture attenuation and material-signature overlap together. (b) The same two quantities against material-signature overlap alone.

Individually, the five degradation axes are not equally damaging (Table 4). At the most severe level tested on each axis alone, overall sensor noise does the most harm (macro-F1 $0.64 \rightarrow 0.39$, mean VOI $\$1012 \rightarrow \490), followed by material-signature overlap (F1 $\rightarrow 0.47$, VOI $\rightarrow \$810$) and dropout probability (F1 $\rightarrow 0.51$, VOI $\rightarrow \$787$); GPR-specific depth attenuation and moisture attenuation, tested over the same relative range, show little standalone effect on either metric (VOI $\rightarrow \$1009$ and $\$1023$ respectively), which is notable given GPR’s importance to raw classification (Section 5.3) and suggests the economic result is more sensitive to generic sensor noise than to this specific channel’s physics. The combined scenario, moving all five axes together, is more damaging than any individual axis: mean VOI falls from $\$1012$ at baseline to $\$985$, $\$868$, $\$359$ and $\$-40$ across increasing combined severity, turning negative before the most severe combined scenario tested. A finer search along the same combined axis locates the point at which mean VOI changes sign at 98% of the way from baseline to the most severe scenario evaluated — a model-specific VOI-zero boundary, not a general claim about when surveying stops being worthwhile: beyond it, this simulation’s economic prediction changes sign. Section 5.11 replaces this one-dimensional severity fraction with an explicit two-dimensional decision-viability map giving the boundary in terms of the underlying degradation parameters.

Degradation axis	Baseline	Value at VOI = 0	Most severe scenario
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Sensor noise	1.0	2.47	2.5
Dropout probability	1.0	3.93	4.0
GPR depth attenuation	1.0	2.47	2.5
Moisture-driven GPR attenuation	1.0	2.47	2.5
Material-signature overlap	0.0	0.59	0.6

Table 5. The combined-degradation VOI-zero boundary (97.7% of the way from baseline to the most severe scenario, Fig. 9a) expressed in the underlying degradation parameters used by the combined-severity sweep, rather than as a bare percentage. Sensor noise and dropout are expressed as multiples of their baseline standard deviation/probability; GPR and moisture attenuation as multiples of their baseline attenuation coefficient; overlap as the fraction each material's response is shrunk toward the six-class mean.

5.9 Classifier choice and probability calibration for the economic layer

The economic layer (Sections 5.4 and 5.7) uses random forest rather than XGBoost, the classifier with the highest raw macro-F1 (Section 5.1), because XGBoost's probabilities are the least well calibrated of the three (Brier 0.070 vs. 0.059 for random forest; Section 5.6), and VOI consumes predicted probabilities directly rather than only the predicted class. This choice is tested, not merely asserted: averaged over 20 independent surveys, mean VOI is \$729 with random forest, \$720 with raw XGBoost (difference \$-9) and \$731 with Platt-calibrated XGBoost (sigmoid calibration, 3-fold; Brier improves to 0.066). All three are within normal sampling variation of one another and all give $P(\text{VOI} > 0) \geq 95\%$: classifier choice for the economic layer does not materially change the economic conclusion in this simulation, which is itself informative — it means the classifier-calibration concern raised by Section 5.6's Brier-score gap, while real, is not what is driving this paper's headline VOI estimate. This comparison uses one fixed batch of 20 survey seeds so the three classifiers are compared on identical draws (a paired design); Section 5.7's canonical baseline VOI pools this same random-forest batch with an independent second batch for a lower-variance point estimate, which is why its value differs slightly from the \$729 quoted here for random forest alone.

5.10 Sensor selection depends on site composition, not price alone

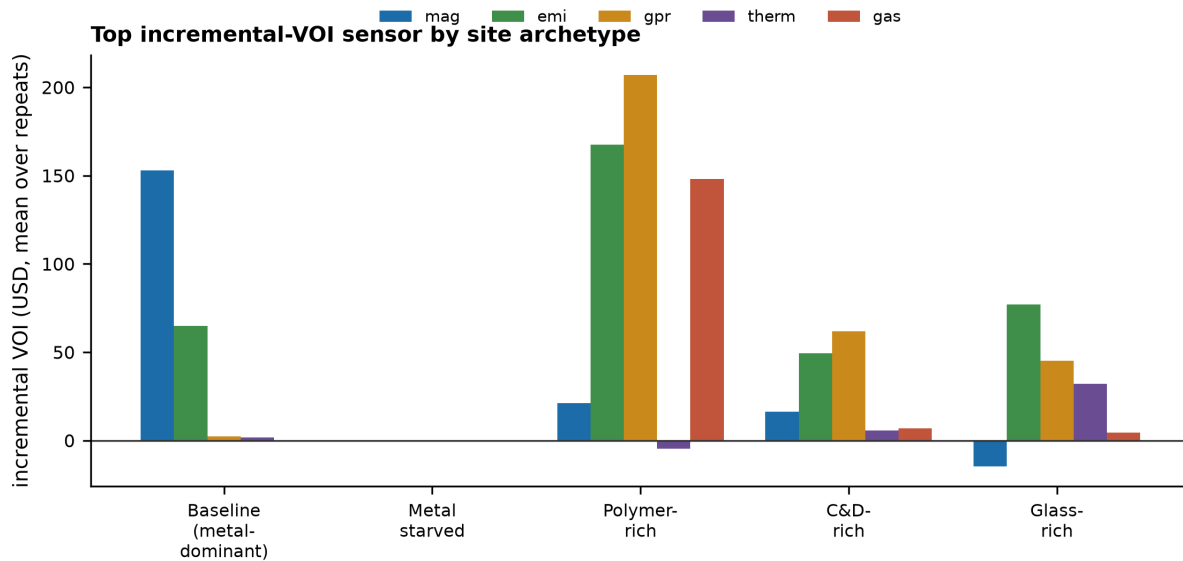


Fig. 10 Incremental VOI (mean over 15 repeats) for each of the five channels, at baseline economics and four deliberately extreme site archetypes: metal devalued to 30% of baseline with no other change; the same metal devaluation plus plastic price and recovery fraction raised 5.0x/4.0x (polymer-rich); plus C&D price/recovery raised 6.0x/2.5x (C&D-rich); plus glass price/recovery raised 8.0x/3.5x (glass-rich).

A grid over metal price (0.5–1.6x baseline) crossed with plastic price (0.4–3.0x baseline), 5×5 points, 15 repeats per point, never changed the winner away from the magnetometer (Fig. 5b's single-point result therefore generalises across ordinary price fluctuation, not just the one price table in Table 2). But deliberately extreme, still economically nameable site archetypes do flip the ranking: starving metal's value alone collapses every sensor's incremental VOI toward zero (no single channel matters when no material is worth much); combined with a hypothetical high-value, high-recovery plastics market, GPR becomes most valuable (\$207), ahead of EMI (\$168) and gas (\$148); with a high-value C&D market, GPR again leads (\$62); with a high-value glass market, EMI leads (\$77). The classification-optimal sensor (GPR, Section 5.3) and the economic-optimal sensor are therefore not fixed properties of the sensing architecture: they are conditional on the site's resource-value structure, and the general claim this paper supports is that sensor suites should be selected against site-specific decision value, not against a generic classification benchmark or a single assumed price table.

5.11 A two-dimensional decision-viability map

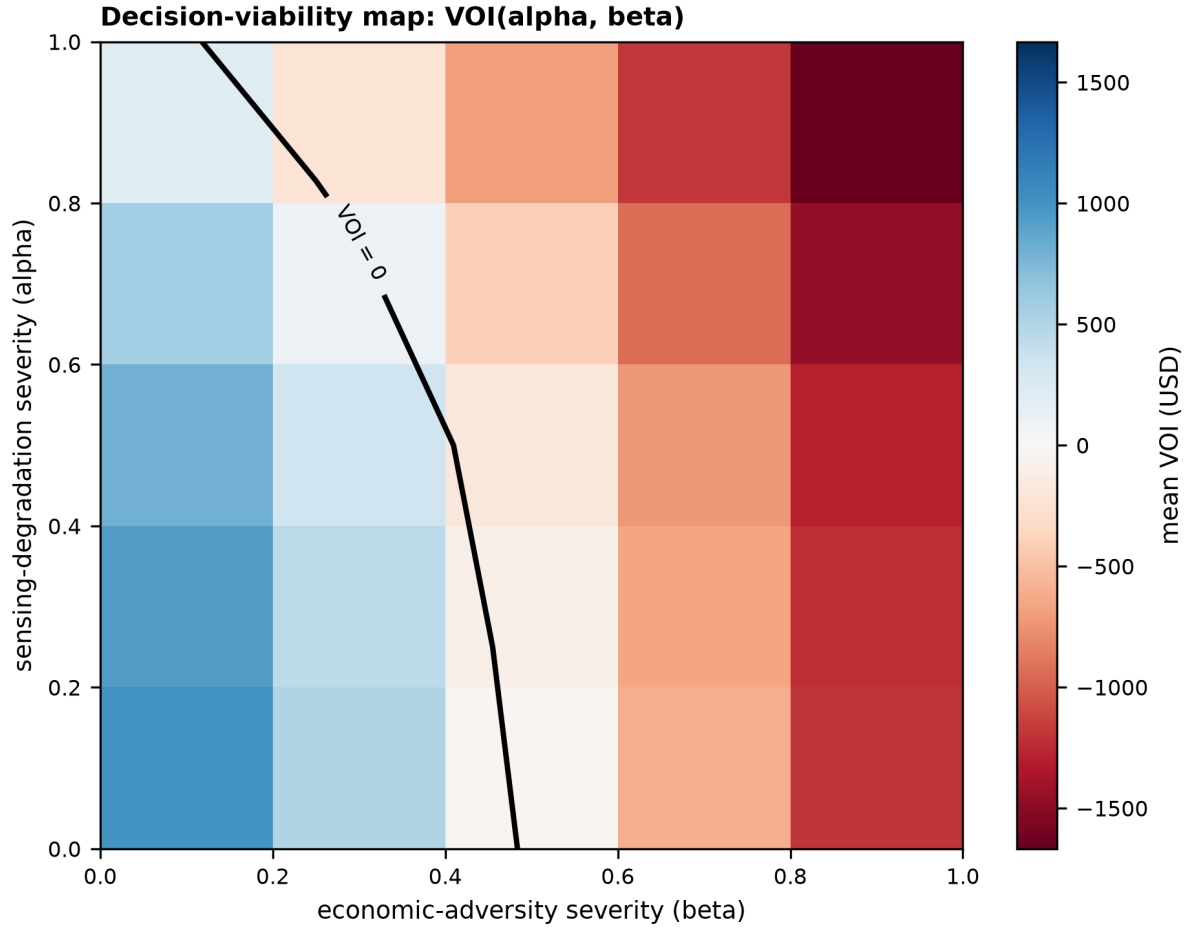


Fig. 11 Mean VOI as a function of sensing-degradation severity (alpha, the same combined parametrisation as Fig. 9a) and economic-adversity severity (beta: secondary-material prices shrunk toward zero and survey cost increased), 5×5 grid, 10 repeats per point, with the $\text{VOI} = 0$ contour drawn in black.

The one-dimensional combined-degradation sweep (Fig. 9a, Table 5) is the $\beta = 0$ slice of this surface. Moving along that slice alone, the VOI-zero boundary sits at 98% combined severity; but the boundary is far more sensitive to economic adversity than to sensing degradation: at $\alpha = 0$ (no added sensing degradation at all), mean VOI already crosses zero at $\beta \approx 0.48$, and at $\alpha = 1$ (the most severe sensing degradation tested) it crosses at $\beta \approx 0.12$ — a comparatively small shift. The decision-viability region $\mathcal{L}_+ = \{(\alpha, \beta): \text{VOI}(\alpha, \beta) > 0\}$ in this simulation is therefore bounded mainly by whether secondary-material markets and survey costs stay favourable, not by how well the sensors themselves perform; a site operator worried about the economics of this approach should stress-test market assumptions before stress-testing sensor specifications.

6 Discussion

6.1 Classification accuracy is not the right sensor-value metric, and the right sensor is not fixed

Section 5.4's central finding — that GPR is the strongest classification contributor but only the magnetometer shows reliable economic value at baseline economics — argues against sizing a multimodal survey by F1 or macro-F1 alone. Section 5.10 goes further: the magnetometer's advantage survives ordinary price fluctuation but not deliberate changes in what the site actually contains, and GPR or EMI become the economically optimal channel under polymer-rich, debris-rich or glass-rich archetypes. The general, falsifiable claim this paper supports is therefore that predictive utility is not decision utility, and that the economically optimal sensor is a function of site composition, market economics and sensor uncertainty jointly — not “magnetometers are economically most useful,” which is only the baseline-economics special case. A sensor package for this application should be chosen, and priced, against its contribution to the value-of-information calculation for the specific material-price structure and composition of the site, not against a generic classification benchmark or a single assumed price table. This is a design implication the original proposal's Stage 1 testing, reporting only F1 scores under one assumed economic table, could not have surfaced.

6.2 What the spatial-validation null result does and does not show

We tested, rather than assumed, whether random splitting overestimates performance for this system and found no detectable effect (Section 5.2). This should not be read as evidence that spatial blocking is unnecessary in general; the mechanism the literature documents [7, 9] depends on a model exploiting local spatial proximity, and our feature set (per-cell sensor readings plus raw coordinates) evidently gives XGBoost, random forest and logistic regression little opportunity to do so. A design that adds explicit neighbourhood-averaged features, or a real dataset whose noise correlation structure differs from our assumptions, could behave differently, and spatial blocking costs nothing to keep as standard practice.

6.3 The plastic/glass limitation arises from the assumed sensor-response physics, not classifier tuning

The proposal's diagnosis — that plastic and glass are weakly magnetic and poorly distinguished by this channel set, and that no amount of classifier tuning fixes this — is supported by our results: three different classifier families all fail on these two classes by a similar margin (Table 3), and adding a channel with distinct plastic/glass responses (NIR) improves plastic specifically without improving glass (Section 5.5). This points to plastic and glass needing materially different remedies (for example, glass may need a different spectral band or a density-based separation step) rather than one “optical fix” for both.

6.4 Limits of the economic model

The VOI conclusion in Section 5.7 depends on the price and recovery-fraction ranges of Table 2, explicitly labelled illustrative, and on treating the excavate/do-not-excavate decision as a single per-cell threshold rather than a portfolio decision across the whole site. Real secondary-material markets are more volatile than the ranges swept here, and the framework's practical strength, as the source proposal itself notes, is that VOI can be recomputed under site- and market-specific inputs rather than relying on any single value estimate.

6.5 Relation to established landfill-mining economics

The finding that survey-guided decisions can convert an unprofitable blind-excavation baseline into a profitable one is consistent with the broader literature's diagnosis that landfill-mining profitability is highly sensitive to a small number of factors that are, in principle, knowable before excavation [1, 2]. This paper's contribution is to make “how much would it be worth to know” a computable, falsifiable quantity for a specific sensing architecture, rather than a general observation about uncertainty.

6.6 Limitations

Every quantitative result in this paper is a prediction from a synthetic, physically motivated but unvalidated model. Sensor-response coefficients (Table 1), noise levels, dropout rates, spatial correlation lengths, and every economic parameter (Table 2) are stated assumptions, not measurements; the robustness sweep (Section 5.8) shows the central economic conclusion depends on assumptions holding within roughly the first 98% of the degradation range explored. The gas-hazard truth label is constructed from the same underlying model as the gas sensor reading, so the reported hazard performance is an upper bound on what independent ground-truth gas testing would show. The three-depth-layer material field is a simplification of real, continuously heterogeneous stratigraphy. No field, laboratory, or test-bed data were used anywhere in this study.

6.7 Load-bearing assumptions

Four assumptions carry most of the paper's conclusions, and different conclusions lean on different subsets of them, so a failure in one need not undermine the others.

A1 — synthetic sensor signatures approximate real landfill sensor separability. Load-bearing for every classification result (Sections 5.1–5.3, 5.5): the metal/organic/C&D-high, plastic/glass-low F1 pattern and the GPR-dominant ablation ranking all depend on it. If real materials are less separable than modelled, classification F1 (and everything downstream of it) will be lower than reported.

A2 — the material-price/recovery-fraction distributions represent the intended decision environment. Load-bearing for every economic result (Sections 5.4, 5.7–5.11): the magnetometer's

baseline VOI advantage, the break-even survey cost and the decision-viability map's beta axis all depend on it directly. Section 5.10 shows this dependence explicitly by varying it.

A3 — the simulator's spatial structure approximates real landfill heterogeneity. Load-bearing mainly for the spatial-vs-random null result (Section 5.2) and the distance-decay diagnostic; a real site with different spatial correlation in its material field or sensor noise could show a leakage gap this simulation did not.

A4 — predicted probabilities are well enough calibrated for expected-value arithmetic. Load-bearing for the economic layer's use of predicted probabilities rather than only predicted classes; Section 5.9 tests this directly and finds the VOI conclusion is not sensitive to it in this simulation, which weakens, but does not remove, its load-bearing status.

7 Field validation and falsification plan

The predictions above can be confirmed or overturned by a staged test programme. Table 6 lists the measurements, the test bed, and the acceptance criterion for each.

Prediction	Test	Failure criterion
Metal, organic, C&D F1 remain high on real waste	Controlled test bed with known buried materials of each class	F1 below the values in Table 3 by more than the robustness sweep's noise-equivalent margin (Fig. 9)
Plastic/glass remain the hard classes without an optical channel	Same test bed, five-channel configuration	Unexpectedly high plastic/glass F1 would indicate the synthetic physics under-modelled real discriminability
NIR improves plastic but not necessarily glass	Same test bed, six-channel configuration with an added optical/NIR unit	Glass F1 improves as much as plastic (would contradict Section 5.5); or neither improves (would contradict the source proposal's diagnosis)
Sensor economic ranking follows site composition (Section 5.10), not a fixed classification-accuracy ranking	Ablation repeated on field data with the same economic model applied to real prices and the site's actual measured composition	Field outcome inconsistent with simulation: GPR (or another non-magnetometer channel) produces a reliably positive incremental VOI exceeding the magnetometer's on a site the simulation's regime map would have classified as metal-dominant
Spatial vs. random validation gap remains small	Repeat the spatial-block/random-split comparison on field data	A gap appears whose 95% CI excludes zero, indicating the null result of Section 5.2 does not transfer
Gas-hazard recall remains high at a safety operating point	Independent, ground-truthed gas testing (not derived from the same sensor)	Recall below 90% at the chosen operating point
VOI remains positive at the site's actual survey cost	Full field pilot with measured excavation and processing costs and actual secondary-material prices	Measured VOI is negative, or the break-even survey cost is below the field survey's actual cost

Table 6. Field validation and falsification criteria. A pilot at full site scale is proposed only after the controlled test-bed stage has been passed.

7.1 Status of claims and assumptions

Every claim in this paper falls into one of four evidentiary categories; conflating them is the most common way a simulation study overstates itself, so Table 7 states each claim's status explicitly rather than leaving it implicit in the prose.

Claim	Status	Evidence/basis
Random splitting inflates classification performance relative to spatial blocking	Not supported in simulation	95% CI includes zero for LR/RF/XGB (Section 5.2)
GPR contributes most to macro-F1 among the five channels	Simulation result	Leave-one-sensor-out ablation (Section 5.3)
Magnetometer contributes the most reliable incremental VOI at baseline economics	Simulation result, economics-conditional	20-survey incremental-VOI analysis (Section 5.4); shown conditional in Section 5.10
NIR improves plastic classification	Conditional simulation result	Assumed synthetic NIR response coefficients (Table 1), not measured spectra
NIR increases VOI	Not supported	Incremental VOI statistically indistinguishable from zero (Section 5.5)
Baseline median VOI is positive	Model-dependent numerical result	Monte Carlo economic model, stated price/recovery/cost ranges (Section 5.7)
VOI becomes negative under severe combined degradation	Model-dependent boundary	Robustness sweep and 2-D decision-viability map (Sections 5.8, 5.11)
The economically optimal sensor depends on site composition, not just price	Supported within tested simulation	Economic-regime grid and extreme-archetype test (Section 5.10)
Classifier choice for the economic layer does not change the VOI conclusion	Supported within tested simulation	RF vs. raw/calibrated XGBoost comparison (Section 5.9)
Field performance will match these simulated values	Untested prediction	Requires the prospective validation in Table 6

Table 7. Status of claims and assumptions. “Simulation result” means the finding is reproducible from the released code under the stated assumptions; it is not thereby a claim about real landfills, which is what Table 6 exists to test.

8 Conclusions

A multimodal sensor-fusion architecture for pre-excavation landfill characterisation was evaluated by simulation against the questions it raises rather than the claims originally made for it. Metal, organic material and construction and demolition debris are classified with high simulated F1 (up to 0.99) under a spatially valid protocol; plastic and glass are not (F1 0.12–0.32), for reasons of the assumed sensor-response physics rather than classifier choice. Random per-cell splitting did not detectably inflate performance relative to spatial blocking in this configuration, a tested and reported null result rather than an assumption. The paper's central finding is that predictive utility is not decision utility and is not fixed: GPR maximises classification accuracy, but the economically optimal sensor at baseline economics is the magnetometer, and it changes to GPR or EMI under deliberately different, still economically nameable site compositions (Section 5.10) — optimal sensor selection is a function of site composition, market economics and sensor uncertainty, not a fixed property of the sensing architecture. Pooling 40 independent survey realisations under baseline economics, the simulated survey has positive median VOI (\$924) and a break-even cost of \$1451, and this classifier-layer choice was itself tested and found not to matter (Section 5.9). A two-dimensional decision-viability map (Section 5.11) replaces the single combined-degradation boundary with a full VOI(α , β) surface, showing the economic conclusion is

considerably more sensitive to adverse market conditions than to sensor degradation itself. All results are predictions, explicitly labelled by evidentiary status in Table 7, and the field validation plan (Table 6) specifies the measurements that would confirm or overturn them.

Declarations

Funding. This work received no external funding.

Competing interests. The author declares no competing interests.

Ethics approval. Not applicable. No human participants or animals were involved.

Consent to participate/for publication. Not applicable.

Prior dissemination. An earlier, non-quantitative description of this architecture was submitted by the author in response to an industrial open-innovation challenge focused on landfill mining. That submission was not peer reviewed or published and contained none of the models, validation protocol, or results reported here.

Data availability. No field data were generated. All model inputs are stated in the paper and the released configuration file.

Code availability. The synthetic-survey generator, classification and economic models, robustness sweep, decision-viability and sensor-selection maps, and figure generators are openly available at <https://github.com/sandlerleon/landmap-sensor-fusion>. Running `landmap_model.py`, `landmap_econ.py`, `landmap_robustness.py`, `landmap_hazard_extra.py`, `landmap_calib_voi.py`, `landmap_sensor_map.py`, `landmap_sensor_map_extreme.py`, `landmap_voi_map.py` and `make_figures.py` reproduces every number and figure.

Author contributions. L.S. conceived the study, developed the models, performed the analysis and wrote the manuscript.

Use of generative AI. During the preparation of this work the author used Claude (Anthropic) to assist with literature search, implementation of the models and drafting of the text. The author reviewed and edited all content, verified every cited reference against its source, and takes full responsibility for the content of the article.

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